

CS481 Task 4 Final Report

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1 Methods

1.1 Data Inspection

The dataset consists of 256×256 RGB images labeled real or fake (AI-generated), with 122,907 training samples and 15,324 validation samples at a 1.2:1 fake-to-real ratio.

Manual inspection of approximately 1000 samples of each category (fake vs real) revealed multiple patterns a model could potentially exploit for classification. Table 1 describes the patterns.

Real Data	Fake Data
Diverse expressions and emotions	Limited expression diversity
Random objects (hats, microphones, etc)	Few or no random objects
Clear, diverse backgrounds	Blurry or distorted backgrounds
Visible skin imperfections	Unrealistically perfect skin
Traditional clothing sometimes present	Almost no traditional clothing
Wide age/racial/ethnic diversity	Mostly middle-aged, light-skin people
Naturally varying eye positions	Nearly identical eye positions

Table 1: Observed visual differences between real and AI-generated faces.

1.2 Architecture

The architecture chosen for the final evaluation took inspiration from the MesoNet [1] architecture. This architecture involves 4 same output dimension convolutional layers, each followed by batch normalization, ReLU, and a 2-D max pooling layer. The output of the last pooling layer is then flattened and passed through a single-layer MLP feed forward network that outputs 2 raw logits, which are then converted to probability via softmax.

This architecture was chosen as the starting point due to its low computational cost, simplicity and model accuracy in paper (95% detection rate for images).

Two different variations of the basic MesoNet were also tested. MesoNet implementation 2 had a 3-layer MLP instead of a single-layer MLP and MesoNet implementation 3 had an extra convolutional layer along with the 3-layer MLP. The purpose of both variations was to give the model extra capacity to learn complex features.

1.3 Hyperparameter Tuning

Due to limited compute, a design of experiments hyperparameter comparison between models was not possible. Instead, each model of interest was trained over 10 trials, with hyperparameters randomly sampled from the search space described in Table 2. The models were compared based on their best performance across the sweep.

Hyperparameter	Range	Justification
Transformation	basic, jpeg_compression, flip_blur_cj ¹ , random_erasing, all	Recommended by literature [2]
LR	0.0005–0.0015	Find optimal learning rate
Optimizer	Adam, AdamW, RMSprop, AdaGrad, AdaDelta, SGD	Compare optimization methods
Epochs	3–10	Limited by compute resources
Batch size	32 (fixed)	Largest size fitting GPU memory

Table 2: Hyperparameter search space used for random search experiments.

2 Results

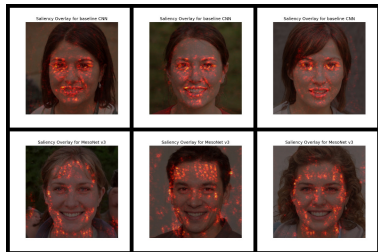


Figure 1: Saliency Map (baseline top, MesoNetV3 bottom)

A hyperparameter analysis showed that top-performing models used an adaptive optimizer, a learning rate around 0.001 ± 0.0001 , and improved with additional epochs. Interestingly, input transformations had little impact, likely because the training data is “easy”, as Table 1 indicates, the visual differences are clear and highly discriminative, explaining high accuracy seen in all models. Saliency analysis (Figure 1) shows that the baseline model primarily looked at lips and eyes while MesoNet V3 looked at eyes and lips too, but also face structure, hair, skin, and the background. See Table 3 for model performance.

Model	Dev Acc.	Description
Dummy Baseline	50.90%	Always predicts the most frequent class.
Baseline CNN	99.09%	Simple CNN adapted from in-class labs
Xception [3]	99.90%	Strong benchmark architecture used widely
MesoNet Basic	99.72%	Basic MesoNet implementation
MesoNet V2	99.94%	MesoNet with 3-layer MLP
MesoNet V3	100%	MesoNet with 3-layer MLP and extra conv layer

Table 3: Model performance results based on dev set.

3 Conclusions

The MesoNet V3 architecture was chosen as the submission model with an Adam optimizer (0.001 LR), flip_blur_cj¹ input transformation with 25 epochs of training due to its high performance and saliency analysis.

4 Contributions

All code and analysis was done by Bogdan Trigubov. Henry Hannawalt helped catch a learning rate bug that caused training instability. Academic works cited on third page.

¹Combination of random horizontal flip, Gaussian blur, and color jitter.

References

- [1] Darius Afchar, Vincent Nozick, Junichi Yamagishi, and Isao Echizen. Mesonet: a compact facial video forgery detection network. In *2018 IEEE International Workshop on Information Forensics and Security (WIFS)*, pages 1–7, 2018.
- [2] Yuhang Lu and Touradj Ebrahimi. A new approach to improve learning-based deepfake detection in realistic conditions, 2022.
- [3] François Chollet. Xception: Deep learning with depthwise separable convolutions, 2017.